

Landmark Extraction Models for Multi-Camera 3D Pose

Survey of the most recent models (2023 - October 2026) for extracting accurate body landmarks from our three fisheye security-camera videos, for offline processing and 3D use.

1. What we need - the selection criteria

criterion	why it matters here
Accuracy first, not speed	the videos are processed offline; hours of GPU time are acceptable
Precise 2D localisation in every view	3D comes from triangulating 2D points of 3 calibrated cameras; 1 px of 2D error at 4 m distance is ~4 mm in 3D, and errors in different views do not cancel
Feet + body joints	the robot needs body and foot joints (heel, toes) now and joint angles later
Robust to our footage	ceiling-mounted fisheye cameras (top-down, distorted), person often seen from behind, a mop handle crossing the body, other people in the room
Consistent over time	the robot follows a trajectory - jitter and left/right swaps between frames are as harmful as a wrong frame
Usable in practice	public code + weights, runs on one GPU (8-24 GB), licence ok for research

2. How these models find landmarks - the four families

family	how it works	what it gives us	examples
A. 2D keypoint detectors (top-down)	a person detector gives a box; the crop is resized; a network outputs, for every joint, a <i>heatmap</i> (probability of the joint at each pixel) or two 1D distributions along x and y (SimCC); the peak is the keypoint and its height the confidence	precise 2D points per camera + confidences, which our triangulation turns into 3D	Sapiens2, CIGPose, RTMW, ViTPose++, YOLO26-pose
B. Multi-view fusion / triangulation	combine the 2D evidence of several cameras into one 3D point: algebraic (weighted DLT, what we have), learned (fusing heatmaps or features across views), or with priors (bone lengths, temporal smoothness)	3D joints in metres; rejection of a wrong view	MEOM, algebraic-priors MV, Kineo, Pose2Sim
C. 3D body-model (mesh) recovery	a network regresses the parameters of a parametric body (SMPL-X or Meta's MHR): joint rotations + body shape + camera; landmarks come from the fitted body	a complete, anatomically valid skeleton with joint angles ; plausible even when joints are hidden	SAM 3D Body, SAM-Body4D, NLF, PromptHMR
D. Multi-view model fitting	optimise one body model so that its projections match the 2D keypoints of all cameras at once, plus smoothness and pose priors	metric 3D + joint angles from all views together	EasyMocap, Pose2Sim (OpenSim IK)

About video: almost all of these models look at **one frame at a time**. Temporal consistency then comes from (a) multi-view triangulation (three cameras must agree), (b) filtering the 3D trajectories, or (c) the few video models (SAM-Body4D, PromptHMR-Vid, the algebraic-priors method) that look at several frames. **About fisheye:** none of the public models is trained on ceiling fisheye images; all are trained on ordinary photos. This is why our pipeline feeds them a **rectified virtual-camera crop** around the person - that turns every model below into a 'fisheye-capable' model. A top-view fisheye dataset (NTOP570K) exists and fine-tuning ViTPose-B on it raised AP by 33 points - an option if the top-down viewpoint still hurts.

3. Family A - 2D keypoint detectors

1. Sapiens2 (Meta)		USE NOW
Facts	ICLR 2026 · vision transformer foundation model pre-trained on 1 billion human images · sizes 0.4B / 0.8B / 1B / 5B · 308 keypoints (body, feet, 40 hand, 243 face) · input 1024 x 768 · Sapiens2 licence (research ok)	
How it finds landmarks	top-down: the person crop goes through a very large ViT at high resolution; a heatmap head outputs 308 heatmaps (UDP decoding, sub-pixel peak); flip-test averages the image and its mirror	
Video	per frame; no temporal model - consistency must come from triangulation and filtering	
Fisheye / our footage	not trained on fisheye; on our rectified crops it gave the most confident and the steadiest keypoints (cam1 test: confidence 0.87, jitter 0.64 px)	
Accuracy	1B: 80.4 mAP, 5B: 82.3 mAP on a 308-keypoint in-the-wild benchmark (RTMW-x: 70.2 on the same set)	
Practical	public weights (no login); 0.8B needs ~2.8 GB in fp16, 0.83 frames/s on an RTX 2070; 1B on a 16 GB+ GPU	
Verdict for this project	Our main model. Best published accuracy for body + feet, already integrated and tested. On the GPU computer, use the 1B model if it has ≥16 GB.	

2. CIGPose (CVPR 2026)		TRY NEXT
Facts	March 2026 · Causal-Intervention Graph network on top of an RTMPose-style backbone · 133 COCO-WholeBody keypoints · Apache-2.0 · MMPose project, checkpoints on Google Drive	
How it finds landmarks	first predicts keypoints with an uncertainty; keypoints that look 'confounded' by the background or occlusion are replaced by learned canonical embeddings, then a hierarchical graph neural network refines all joints together so the skeleton stays anatomically plausible	
Video	per frame	
Fisheye / our footage	not trained on fisheye - use our rectified crops. Its strength (robustness to occlusion and clutter) fits our scenes: mop handle, desks, chairs	
Accuracy	CIGPose-x 67.0 AP whole-body (67.5 with UBody) - new SOTA on COCO-WholeBody among RTMPose-type models	
Practical	needs an MMPose environment (no ONNX yet); medium speed; 384 x 288 input	
Verdict for this project	Best candidate for a second, independent opinion next to Sapiens2 (better than RTMW, same keypoint format). Disagreements between Sapiens2 and CIGPose flag unreliable frames.	

3. RTMW / RTMPose (OpenMMLab, via rtmllib)		USE NOW
Facts	2023-2024 · CNN + SimCC head · 133 COCO-WholeBody keypoints · Apache-2.0 · ONNX, runs without PyTorch	
How it finds landmarks	predicts two 1D distributions per joint (one along x, one along y) instead of a 2D heatmap - fast and sub-pixel	
Video	per frame (rtmllib has a simple tracker)	
Fisheye / our footage	works on our crops; the 'cocktail13' RTMW-x release put the left toes on the head, the 'cocktail14' release is correct; agrees with Sapiens2 within 1-4 px on all joints	
Accuracy	70.1 AP whole-body (384 x 288)	
Practical	~30 frames/s on an RTX 2070; standard 2D backend of Pose2Sim	
Verdict for this project	Already integrated as the fast baseline and cross-check . Not the most accurate.	

4. ViTPose / ViTPose++ (Univ. Sydney)		ALTERNATIVE
Facts	NeurIPS 2022 / TPAMI 2023 · plain vision transformer + simple heatmap decoder · sizes B, L, H, G (1B) · 17 body keypoints (ViTPose++ also animals/whole-body heads) · Apache-2.0 · in Hugging Face transformers	
How it finds landmarks	top-down heatmaps from a plain ViT; multi-dataset training	
Video	per frame	
Fisheye / our footage	rectified crops needed; the top-view fisheye dataset NToP570K was built on ViTPose - a fine-tuned version is the most direct route to a 'ceiling-view' model	
Accuracy	ViTPose-G 80.9 AP on COCO test-dev (body only); ViTPose++-H 78.5 AP	
Practical	easy (transformers), but no foot keypoints in the standard models	
Verdict for this project	Strong body accuracy but no feet ; Sapiens2 is the newer and more complete successor of the same idea. Relevant only if we fine-tune on top-view fisheye data.	

5. YOLO26-pose (Ultralytics)		NOT FOR US
Facts	2026 · single-stage detector that outputs boxes + keypoints in one pass, RLE keypoint loss · 17 body keypoints · AGPL-3.0	
How it finds landmarks	bottom-up/one-stage: every person and keypoint predicted from the full image at once	
Video	per frame + built-in trackers (BoT-SORT/ByteTrack)	
Fisheye / our footage	runs on raw fisheye frames (lower resolution per person, distorted)	
Accuracy	YOLO26x-pose 71.6 AP (COCO body)	
Practical	real-time, very easy	
Verdict for this project	Built for speed, not maximum accuracy, and no foot keypoints . We use the YOLO26x detector only to find and track people.	

4. Family B - multi-view fusion and triangulation

6. MEOM - Multi-view Expected-OKS Maximisation		LATER STAGE
Facts	arXiv, 31 Aug 2026 · newest work on multi-view triangulation · works on 2D heatmaps · code not announced yet	
How it finds landmarks	instead of taking one 2D point per view and intersecting rays (DLT), it places each 3D joint where the probability mass of all views' heatmaps agrees - so a second, hidden candidate in one view (multimodal heatmap under occlusion) is not lost	
Video	per frame	
Accuracy	19.1 mm MPJPE on Human3.6M with 3D supervision, better than volumetric methods at half the cost; strong on occluded data without 3D labels	
Practical	needs heatmaps from the 2D model (Sapiens2 produces them) and calibrated cameras	
Verdict for this project	The natural upgrade of our DLT triangulation once the basic 3D pipeline runs; our backend can keep the heatmaps for this.	

7. Unconstrained Multi-view Pose with Algebraic Priors		ALTERNATIVE
Facts	arXiv, Apr 2026 · uncalibrated multi-view, transformer-based triangulation + Gröbner-basis geometry corrector + temporal equivariant rectifier · code not announced	
How it finds landmarks	fuses 2D keypoint tokens from all views with a transformer (no camera parameters), then enforces projective-geometry constraints and temporal coherence over the sequence	
Video	yes - temporal module across frames	
Accuracy	state of the art for uncalibrated multi-view; closes much of the gap to calibrated methods	
Practical	research code status unclear	
Verdict for this project	Interesting because it needs no extrinsic calibration , but calibrated triangulation stays more accurate - we have (or will have) calibration.	

8. Kineo - calibration-free metric motion capture		ALTERNATIVE
Facts	arXiv Oct 2025 · open code (CC BY-NC-SA 4.0) · sparse, unsynchronised and uncalibrated consumer cameras	
How it finds landmarks	off-the-shelf 2D keypoints + dense point maps; a graph-based global optimisation jointly estimates camera poses, lens distortion (Brown-Conrady), time offsets and metric 3D keypoints	
Video	yes - uses spatio-temporal keypoint sampling over the whole sequence	
Accuracy	83-91 % lower world joint error than earlier calibration-free methods	
Practical	processes 80 min of footage in ~36 min; designed for normal lenses (our fisheye would need our intrinsics or rectified views)	
Verdict for this project	A backup for our two open steps (camera positions and time sync): it solves both from the person's motion. Also useful to cross-check our own extrinsics.	

9. Pose2Sim		LATER STAGE
Facts	open-source pipeline (BSD-3), actively maintained, peer-reviewed accuracy studies · RTMPose 2D + weighted DLT + filtering + OpenSim inverse kinematics	
How it finds landmarks	2D per camera (RTMPose/RTMW), robust weighted triangulation with view rejection, Butterworth filtering, then a biomechanical skeleton is fitted (OpenSim) to get joint angles	
Video	temporal filtering of 3D trajectories; synchronisation tool included	
Accuracy	joint angles within ~2-6° of marker-based motion capture in validation studies	
Practical	fisheye supported through its calibration; our framework already follows the same design	
Verdict for this project	The reference design for our joint-angle stage : we can export our 3D landmarks to its OpenSim kinematics step to get anatomically valid joint angles for the robot.	

5. Family C/D - 3D body models and multi-view fitting

10. SAM 3D Body (Meta)		LATER STAGE
Facts	CVPR 2026 · single-image full-body mesh recovery · Momentum Human Rig (MHR) body model, separate skeleton and shape · trained on 7 M annotated images · SAM licence	
How it finds landmarks	an encoder-decoder (DINOv3 / ViT-H backbone) predicts the MHR pose and shape; a separate hand decoder refines hands; it can be prompted with 2D keypoints or a mask to fix ambiguous cases or pick the right person	
Video	per image (see SAM-Body4D for video); a multi-view extension triangulates its 2D keypoints	
Fisheye / our footage	use rectified crops; prompting it with our triangulated/projected keypoints keeps all views consistent	
Accuracy	state of the art single-view mesh recovery, strong generalisation in the wild	
Practical	public code + checkpoints (access request on Hugging Face); large model	
Verdict for this project	Best candidate for joint angles and a full skeleton : fit MHR to our multi-view keypoints, or use its per-view estimate as a prior when joints are hidden.	

11. SAM-Body4D		ALTERNATIVE
Facts	arXiv Dec 2025 · training-free video extension of SAM 3D Body · open code	
How it finds landmarks	runs SAM 3D Body on identity-consistent segmentation masks across the video; an occlusion-aware module recovers hidden body parts from neighbouring frames	
Video	yes - temporally stable mesh trajectories	
Accuracy	more stable than per-frame SAM 3D Body on in-the-wild videos	
Practical	heavy; single-view - depth/scale less reliable than our triangulation	
Verdict for this project	Good for temporal smoothness and occlusion ; single-camera depth makes it a prior, not the final 3D measurement.	

12. NLF - Neural Localizer Fields (MPI Informatics)		ALTERNATIVE
Facts	NeurIPS 2024 · 3D pose and shape for any point of the body (any skeleton, any surface point) · PyTorch and TensorFlow models, research licence	
How it finds landmarks	a network outputs, for any queried body point, a 3D heatmap localiser - so one model gives SMPL joints, COCO joints, biomechanical marker positions, or mesh vertices	
Video	per frame (batched)	
Fisheye / our footage	accepts the camera intrinsics, so it works on our rectified crops; yields metric 3D in the camera frame	
Accuracy	outperformed the state of the art on several 3D benchmarks at publication	
Practical	public weights; flexible skeleton definition is ideal for a robot kinematic model	
Verdict for this project	Unique ability to output exactly the joints the robot needs (e.g. anatomical joint centres); useful as a 3D prior or to cross-check our triangulated skeleton.	

13. PromptHMR (CVPR 2025) / PromptHMR-Vid		ALTERNATIVE
Facts	promptable SMPL-X mesh recovery · video version with temporal transformer + SLAM for world coordinates	
How it finds landmarks	image + prompts (boxes, masks, text, interaction) → SMPL-X; the video version adds a temporal transformer for smooth motion	
Video	yes	
Accuracy	3DPW PA-MPJPE 35.5 mm (video version)	
Practical	single-camera; SMPL-X licence	
Verdict for this project	Strong monocular video model, but our three calibrated cameras give real metric 3D; at most a prior.	

14. EasyMocap (Zhejiang University)		LATER STAGE
Facts	mature open-source toolbox · fits SMPL / SMPL+H / SMPL-X / MANO to 1-23 calibrated, synchronised cameras	
How it finds landmarks	2D keypoints from any detector, then one body model optimised against all views with pose priors and temporal smoothness	
Video	yes - temporal smoothing terms	
Accuracy	multi-view constraints make it metric and robust; quality depends on the 2D input	
Practical	needs our calibration converted to its format; fisheye via rectified/undistorted keypoints	
Verdict for this project	Ready-made way to get a full body model with joint angles from our multi-view keypoints.	

15. Top-view fisheye fine-tuning (NToP570K dataset)		ALTERNATIVE
Facts	2024-2025 · 570 k synthetic top-view fisheye images rendered with NeRF, with 2D and 3D labels	
How it finds landmarks	not a new model: a dataset to fine-tune existing 2D/3D models (ViTPose, HybriK) to the ceiling-camera viewpoint	
Accuracy	fine-tuning ViTPose-B: +33.3 AP on top-view fisheye; HybriK: -53.7 mm PA-MPJPE	
Practical	requires training	
Verdict for this project	Only if our rectified crops still fail when the person is directly below a camera (strong top-down view).	

6. Comparison

#	model	year	type	keypoints	video	3D use	verdict
1	Sapiens2	2026	2D heatmap ViT	308 (body, feet)	frame	triangulate	use now (main)
2	CIGPose	2026	2D + graph net	133	frame	triangulate	try next
3	RTMW	2024	2D SimCC	133	frame	triangulate	use now (baseline)
4	ViTPose++	2023	2D heatmap ViT	17	frame	triangulate	alternative
5	YOLO26-pose	2026	1-stage 2D	17	frame+track	triangulate	detector only
6	MEOM	2026	heatmap fusion	any	frame	3D directly	later stage
7	Algebraic priors	2026	uncalibrated MV	any	yes	3D directly	alternative
8	Kineo	2025	calib-free MV	any	yes	3D + calib + sync	alternative / backup
9	Pose2Sim	2022-26	MV pipeline + IK	RTMPose	filter	3D + angles	later stage
10	SAM 3D Body	2026	mesh (MHR)	full body	frame	prior / fit	later stage
11	SAM-Body4D	2025	video mesh	full body	yes	prior	alternative
12	NLF	2024	3D any point	any	frame	camera-space 3D	alternative
13	PromptHMR-Vid	2025	video SMPL-X	SMPL-X	yes	prior	alternative
14	EasyMocap	2021-25	MV body fitting	SMPL-X	yes	3D + angles	later stage
15	NToP570K	2024-25	fine-tune data	-	-	-	if needed

Accuracy numbers come from different benchmarks (COCO body, COCO-WholeBody, Sapiens' 308-keypoint set, Human3.6M, 3DPW) and are **not directly comparable** across rows; the decisive test for our project is the multi-view reprojection error and bone-length stability on our own footage.

7. Recommended accuracy-first pipeline

stage	choice	reason
2D landmarks	Sapiens2 (1B on a ≥ 16 GB GPU, else 0.8B) on rectified virtual-camera crops	highest published accuracy with feet; already integrated and tested
second opinion	CIGPose-x (replacing RTMW as cross-check)	independent, occlusion-robust, more accurate than RTMW; disagreement marks bad frames
3D	weighted DLT with view rejection (implemented) → later MEOM-style heatmap fusion	uses all three calibrated views; heatmap fusion keeps occluded alternatives
calibration + sync	reference object + our tools; Kineo as cross-check / backup	Kineo estimates camera poses and time offsets from the person's motion
time consistency	3D filtering; optional SAM-Body4D prior	robot needs smooth trajectories
joint angles	SAM 3D Body / MHR or Pose2Sim-OpenSim skeleton fit to the 3D landmarks	anatomically valid angles with constant bone lengths

Sources

Sapiens2: arxiv.org/abs/2604.21681, github.com/facebookresearch/sapiens2 · CIGPose: arxiv.org/abs/2603.09418, github.com/53mins/CIGPose · RTMW: arxiv.org/abs/2407.08634, github.com/Tau-J/rtmlib · ViTPose/ViTPose++: arxiv.org/abs/2204.12484, arxiv.org/abs/2212.04246 · YOLO26: docs.ultralytics.com/models/yolo26 · MEOM: arxiv.org/abs/2608.30521 · Algebraic priors: arxiv.org/abs/2604.24312 · Kineo: arxiv.org/abs/2510.24464 · Pose2Sim: github.com/perfanalytics/pose2sim · SAM 3D Body: arxiv.org/abs/2602.15989, github.com/facebookresearch/sam-3d-body · SAM-Body4D: arxiv.org/abs/2512.08406 · NLF: arxiv.org/abs/2407.07532, github.com/isarandi/nlf · PromptHMR: arxiv.org/abs/2504.06397 · EasyMocap: github.com/zju3dv/EasyMocap · NToP570K / top-view fisheye: arxiv.org/abs/2402.18196